METHODOLOGY NOTE FOR DISCUSSION WITH HEALTHCARE PROFESSIONALS

# How OUCH calculates its figures

Formulas, thresholds, sources and limits of each indicator in the report, so that you can check and challenge them.

**Reference document**
5 November 2026
OUCH v0.11.0

## WHAT THIS NOTE DESCRIBES

OUCH is a pain diary that the person fills in themselves, in under a minute a day. Its report summarises that diary with a few calculations: means, breakdown of days, comparison with the previous period, flares, before-and-after comparisons around a treatment change, correlations. For each one, this note gives the formula, the threshold, its source and its limit.

## HOW TO READ IT

The data is self-reported and nothing is interpreted: no diagnosis, no treatment advice. Conditions, “diagnosed” or “suspected”, are declared by the patient and reproduced as they are. References, numbered in square brackets as in the report, are listed at the end of the note. When a choice is specific to OUCH and not consensual, such as the definition of flares, the note says so.

## PRINCIPLES THAT APPLY TO EVERY CALCULATION

- ✓ Use only the days with an entry. A missing day is excluded, never estimated or filled in.
- ✓ State, with each figure, the number of days it rests on.
- ✓ Cite the source of each threshold and say when it is extrapolated.

## WHAT THE REPORT DOES NOT DO

- × Make a diagnosis or advise a treatment.
- × Correct correlations for autocorrelation or multiple comparisons, or compute a p-value.
- × Present an association as a cause.
- × Send the data anywhere: everything is computed on the person’s device.

## 1 · DATA COLLECTED

| DATA | SCALE | REMARK |
|---|---|---|
| **Pain of the day** | integer from 0 to 10 | The only measure needed for a day to count. Numeric scale, a recommended intensity measure[1][3] and widely validated[10]. Rated for the whole day, preferably in the evening, as the screen says: an evening rating captures the day’s average level well[20][22], with a slight weight on the end of the day[21], and 24-hour recall limits memory bias[6]. Slider, or eleven boxes on flare days: same scale. |
| **Fatigue, brain fog, stress** | 0 to 10, ↑ = more discomfort | Simple numeric scales, **not validated**. |
| **Sleep quality, mood, activity** | 0 to 10, ↑ = better | Simple scales, **not validated**. Sleep quality entered on a given day describes the previous night. Sleep duration, in hours, is self-reported. |
| **Painful areas** | 25 self-reported areas, left and right | The 19 WPI regions[16][17], plus head, hands, feet, pelvis: not a WPI score (§ 10). |
| **Treatments** | register + the day’s doses | Name, type (regular, as needed, unspecified), dose and dates of each dosage. Each day: doses, relief (none to major) and side effects, for as-needed treatments. |
| **Weather, “what helped” actions** | measurements; free labels | Optional weather (Open-Meteo, the person’s location; mean pressure of the day). Actions: walking, warm bath… counted, not ranked. |

## 2 · PERIODS AND COMPLETENESS

| NOTION | DEFINITION |
|---|---|
| **Analysed period** | From the date of the last appointment of the same kind (family doctor or pain clinic) to the report date, both included: `N = (report date − appointment) + 1` days. |
| **Previous period** | Between the last two appointments of the same kind if they are 7 to 366 days apart (length sometimes differs from **N**, and is flagged). Otherwise: the **N** days before the appointment. All comparisons in the report are made against this baseline. |
| **Days with an entry** | Number of days with a pain level, shown as “x / N” with its percentage, for both periods. A day where only an area, a note or a treatment is entered is saved but not counted; a day without an entry is never counted as “no pain”. |
| **Completeness by dimension** | For each dimension (fatigue, sleep, areas, treatments…), the share of days with an entry on which it is recorded. Dimensions never entered are not listed. |
| **Days written afterwards** | Entry created after 04:00 the day after the day concerned. The report counts days written **two or more days** afterwards (the day before is well recalled[6], although one study finds recall at 24-48 h unreliable[22]) and writes a sentence only if there are any; they stay in every calculation. |

## 3 · PAIN STATISTICS

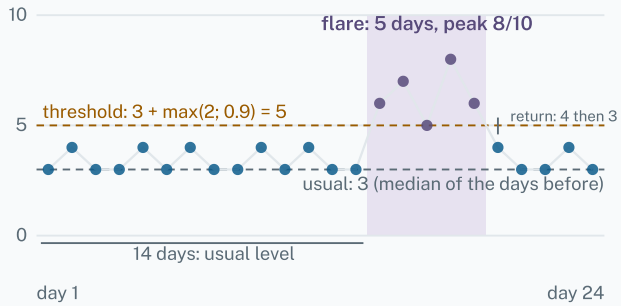
| INDICATOR | CALCULATION | DETAILS |
|---|---|---|
| **Mean and standard deviation** | Arithmetic mean of the days with an entry; sample standard deviation, $\sqrt{(\Sigma(x - \text{mean})^2 / (n - 1))}$ . | Standard deviation set to 0 if there are fewer than 2 days. |
| **Median and quartiles [Q1–Q3]** | Linear interpolation between the sorted values: position $(n - 1) \times p$ , like R’s default type 7. | The median of integers can be x.5. |
| **Minimum, maximum** | Extremes of the days with an entry. | |
| **Mild, moderate, severe days** | Share of days ≤ 3, from 4 to 6, ≥ 7. The number of severe days is this share × n, rounded. | Conventional thresholds. The Boonstra study[5] (2,854 patients) shows they vary with the impact on daily life: mild ≤ 5, moderate 6–7, severe ≥ 8 on average, but ≤ 3, 4–6, ≥ 7 among those who catastrophise little, the case used here. Benchmarks, not a diagnosis. |
| **0–10 distribution** | Share of days at each level, for both periods (histogram). | |
| **Relative change** | `(period mean − previous mean) / previous mean` , computed if a previous period exists and its mean is above 0. | The report states that the change “reaches” the 30% threshold if its absolute value is ≥ 30%, “stays below” it otherwise. A benchmark drawn from group studies of improvement under treatment[2][4][8][9]: it indicates neither an effect nor what matters to the person. Only the relative criterion is tested, not the 2-point one. |
| **7-day rolling mean** | For each day on the chart, the mean of that day’s value and the 6 preceding calendar days, shown if at least 4 of those days have an entry. | Smooths day-of-week effects. The points on the chart remain the raw values. |
| **Week-by-week table** | Blocks of 7 days from the start of the period (the last is shortened): days with an entry, mean pain and doses. | Detailed version only. |
| **Day of the week** | Difference between each weekday’s mean pain and the period’s, if each has ≥ 4 days. Band: $\pm 2.69 \times \text{standard deviation} / \sqrt{n_{\text{min}}}$ | 2.69: 5% for seven comparisons (Bonferroni). Days are not independent: indicative only. |

## 4 · FLARE DETECTION

A flare is a run of days clearly above what is **usual for the person**, not above a fixed value: 7/10 is an ordinary day for one person and an emergency for another. **There is no consensus definition of a flare**; this one is specific to OUCH. It reuses the minimal clinically important difference[2][4], established for improvement and applied here to a worsening, which is an extrapolation. **It is not looked for in someone who follows only migraine**: a migraine comes in attacks, with pain-free days between, and reading "above the usual" would say nothing true; the report then counts headache days instead (§ 11). It is computed over the whole history, because a day's usual level needs entries from before the report period. An episode is attributed to the period in which it starts; its days are counted where they fall.

| PARAMETER | VALUE | ROLE |
|---|---|---|
| **Usual level** | Median pain of the previous 28 days, excluding days already in a flare | The median is little affected by extreme days. At least 10 days with an entry are needed in this window; otherwise nothing is detected and the report says "too few days with an entry". |
| **Flare threshold** | usual + max(2 points; 30% × usual) | Above a usual level of about 7.7/10, the threshold exceeds 10: the person cannot be detected this way, which is better than an invented threshold. |
| **Floor** | 4/10 | To open an episode, pain must also reach 4. |
| **Minimum length** | 3 consecutive days above the threshold | A day without an entry between two high days does not break the run. A shorter episode is discarded. |
| **Return to usual** | pain ≤ usual + 1 | A day in between resets the count of return days. |
| **End of an episode** | 2 days with an entry back at the usual level | A gap of more than 3 days between two entries closes the episode with an "unknown" end. The last day of an episode is its last day above the threshold. |
| **Ongoing episode** | last high day at most 1 day before the report date, with no return | Flagged as such in the report. |

**Worked example.** Fourteen days of pain between 3 and 4: the usual level is the median, 3, and the threshold 3 + max(2; 0.9) = 5 . The next five days are 6, 7, 5, 8 and 6, all ≥ 5: a 5-day episode, peak 8/10. Then a 4 (≤ 3 + 1: back to usual) and a 3: two return days, the episode is closed, with a return in 1 day. This case is in the software's tests.




### What comes before a flare (detailed version)

For each flare, the report gives the **mean of the 3 previous days** for six measures (sleep duration and quality, stress, fatigue, brain fog, mood, with at least 2 days with an entry), and below it the **usual mean of the 28 days before**, excluding flares (at least 10 days). Treatment changes noted in the previous 14 days are added. Nothing is highlighted: in a trial where anything more than one standard deviation away was flagged, something was flagged before 37% of the 3-day windows that precede no flare, and the reader would see a cause in it.

## 5 · TREATMENTS

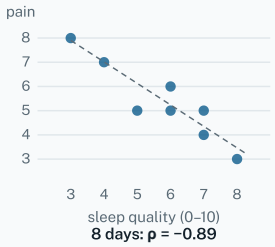
| INDICATOR | CALCULATION |
|---|---|
| **Types** | **Regular**: taken every day, ticked by default, the person unticks a missed dose. **As needed**: number of doses noted, with a prescribed daily maximum. **Unspecified**: treatment added without a description. Each dose change closes a dosage period and opens another. |
| **Days taken** | Days with at least one dose (number of doses absent or > 0). |
| **Self-reported adherence (regular)** | `days taken / (days taken + days explicitly marked missed)` . Days without an entry are left out of the calculation. |
| **Doses (as needed)** | Sum of doses; maximum in one day, flagged if it exceeds the prescribed maximum. Side effects: each effect reported, with number of days and first and last dates. |
| **Relief (as needed)** | Count of doses by level; in the short version, the share of rated doses judged “moderate or major”: `(moderate + major) / rated doses` . Not asked for regular treatments. |
| **Rescue-medication days** | Days with at least one dose of an “as needed” or “unspecified” treatment, out of the days with an entry. Same calculation within each flare. |

Limit specific to these calculations: since a regular treatment is ticked by default, a forgotten dose that was not unticked counts as taken, so self-reported adherence may be overestimated.

## 6 · FACTORS ASSOCIATED WITH PAIN (DETAILED VERSION)

**Spearman correlation** (ρ) between a factor and the pain of the **same day**, over both periods combined: it is the Pearson correlation computed on ranks, ties receiving their mean rank. It assumes neither a straight line nor a normal distribution and suits 0–10 scales. A factor is shown only from **14 pairs** of values, and not if either of the two is constant.

**Example.** Sleep (quality) 4, 6, 6, 7, 8, 5, 3, 7 and pain 7, 5, 6, 4, 3, 5, 8, 5 over eight days. Sleep ranks: 2, 4.5, 4.5, 6.5, 8, 3, 1, 6.5; pain ranks: 7, 4, 6, 2, 1, 4, 8, 4. The correlation of these two lists of ranks gives **ρ = −0.89**: better nights go with lower pain, over these eight days. Dotted line: a visual guide only.




| FACTOR | VALUE PAIRED WITH THE DAY’S PAIN |
|---|---|
| **Sleep (quality, duration)** | Value entered that day, which describes the previous night. The link runs both ways, more often sleep → pain[14]. |
| **Stress, fatigue** | Value of the day. Fatigue is a co-symptom, not a factor. |
| **Previous day’s activity** | Activity of the previous calendar day (an entry is needed that day). Too much and too little activity are both associated with more pain[15]: a monotonic correlation can therefore miss it. |
| **Positive actions** | Number of actions noted that day. The link may run both ways. |
| **Atmospheric pressure** | Change in hPa since the last day with an entry. Inconsistent literature[7][11][12][13]. |

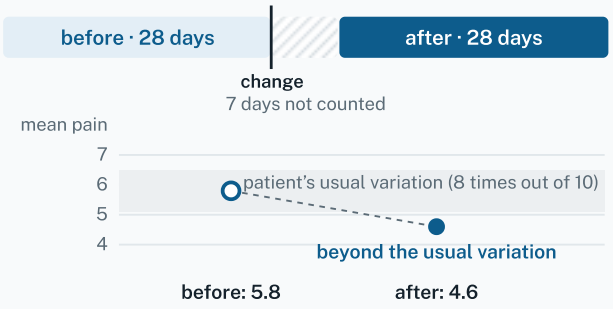
**Strength.** |ρ| < 0.1: negligible; < 0.3: weak; < 0.5: moderate; above: strong (conventional labels, with no test, p-value or confidence interval). **Median comparisons** (at least 3 days per group): the day after a very active day (previous day’s activity ≥ 8); at least 2 positive actions versus none; period days versus the others, if the person includes them.

## 7 · BEFORE AND AFTER A CHANGE OF REGULAR TREATMENT

New in version 0.3.0 (the other calculations are unchanged since 0.2.1, apart from the day-of-week table, added in 0.4.0, section 3). For each start, dose change or stop of a **regular** treatment, the report (both versions, for the period and the 5 weeks before it) and the Trends screen compare the diary before and after. It is a description of two periods: it does not say whether the treatment "works". As-needed treatments have no comparison: what they do is itself a measure.

| PARAMETER | DEFINITION |
|---|---|
| **Periods compared** | Date of the change: start or new dose, its first day; stop, the day after the last day taken. **Before**: the 28 days before. **Settling in**: the following 7 days, not counted (a regular treatment rarely acts on the first day). **After**: the following 28 days. 14 days with an entry are needed on each side, otherwise no figure; their number (before · after) is always shown. Until "after" has elapsed, the comparison is "in progress". |
| **Measures** | Mean pain; flare days out of 28; as-needed doses per day with an entry (a dose without a number counts as one). Flagged under "Good to know", never corrected: another regular-treatment change in the period, or a change made during a flare. |

**Reading the comparison.** The settling-in week is set aside. At the bottom, mean pain before and after is placed next to the patient's usual variation. Fictitious figures: here the drop falls outside the band, which only says it is not ordinary for them, not that the treatment works.




### The patient's usual variation

The diary is cut into 28-day periods from the first day with an entry (at least 14 days with an entry in each, excluding those that overlap the comparison). The usual variation is the **80$^{th}$ percentile** (type 7) of the absolute differences between consecutive periods, with **at least 6 pairs**. The report says "8 differences out of 10 between two consecutive periods are below X points".

**Limits.** One comparison per change, with no test. An improvement may follow a treatment started at the bottom of a painful period (regression to the mean), the weather, the season or another change.

## 8 · WHERE THE METHOD IS SOLID, WHERE IT IS LESS SO

### 1. Solid

Standard statistics, formulas published here and tested: 558 automated tests pass on this version, before-and-after comparisons included.

Validated 0–10 numeric scale, 24-hour recall, missing days never imputed.

### 2. More fragile

Flare definition, extrapolated and non-consensual. Before-and-after comparisons: no test, exposed to regression to the mean. Mild, moderate, severe categories are conventional. Fatigue, sleep and mood scales (and § 11's) not validated.

Seven correlations read without correction for multiple comparisons, on non-independent days.

Days written afterwards: recall is less reliable; they are counted in the report, never set aside.

### 3. Missing

No clinical validation of OUCH itself: no comparison with a validated instrument (BPI, for example), and no flares confirmed by patients or clinicians.

Missing days are not necessarily random: people sometimes fill in less on bad days. Nothing is verified by a clinician.

## 9 · WHAT THE PATIENT NOTICED (REPORT)

In both versions of the report, a section carries the leads the patient **chose to keep** for the appointment (appendix, "Trends: leads"), recomputed over the two periods of the report with the same rule. Each line gives the mean pain and the number of days of the two groups compared. **One-week trials** that ended in these periods are added, when each kind of day counts at least 2; the report reminds that seven days is little. A kept lead that no longer passes the rule on the report's data is left out. Nothing in it is interpreted: these are the patient's observations, unverified, **with no cause established**. Limits: the Trends leads are read without correction for multiple comparisons, on non-independent days, and the patient keeps only the ones that speak to them: it is a selection. The actions in trials are the ones they noted themselves (free text); OUCH suggests none.

## 10 · WHERE THE PAIN IS FELT (REPORT AND TRENDS)

| INDICATOR | CALCULATION | NOTES |
|---|---|---|
| **Areas reported** | Share of logged days on which each area is reported, most frequent first (a day with no area entered reports none); front and back silhouette tinted (detailed version). | 25 areas: the 19 WPI regions[16][17] with the side, plus head, hands, feet and pelvis. Areas from before the side ("Arms") stay as they were. |
| **WPI regions reported** | WPI regions (out of 19) reported on at least one day; average on the days when at least one is; days of "pain everywhere". | A count of what is declared, day by day: not the WPI (past week), nor a score, nor a diagnostic criterion. Head, hands, feet, pelvis, "everywhere" and old areas are not among them. |
| **Trends: areas** | Silhouette and count over the chosen period. **Flares**: areas on flare days against the other days (≥ 6 and ≥ 15 days); an area is named with ≥ 3 flare days and a 25-point gap. **Weeks**: the 6 most reported areas, over 13 weeks. | Descriptive: the days of one flare are not independent, no cause suggested. Cautious, unsourced thresholds; the grid has none. |

## TEN QUESTIONS FOR YOU

1. Does the definition of flares (at least 3 days more than 2 points or 30% above the 28-day median) make sense to you? Which criterion would you prefer?
2. Are the categories ≤ 3, 4–6, ≥ 7 useful to you, or do you read the person's course against their own baseline?
3. Are a 7-day mean and a period compared with the previous one enough, or are other summaries needed?
4. Are correlations useful in consultation, or are they at risk of being over-interpreted? Should they be removed from the general practitioner version?
5. Which indicators do you really use, and which would you remove from the report?
6. What data do you lack today to make decisions, that a diary could provide?
7. Is the before-and-after comparison around a treatment change (28 days before, 7 days not counted, 28 days after, with the patient's usual variation) a useful format? What length would you choose?
8. Is the "What the patient noticed" section (kept leads, one-week trials) useful to you, or might it steer the appointment towards a cause nothing establishes? Should it be removed from the general practitioner version?
9. Is the split into 25 areas (the 19 WPI regions with the side, plus head, hands, feet and pelvis) useful to you? Should there be an intensity per area, or another split?
10. Do the labels, bands and illness profiles of the indicators in § 11 seem right to you? Which sensitive points (bowel, urinary, fever) should be changed or removed?

## 11 · OTHER INDICATORS NOTED (REPORT AND TRENDS)

These indicators are **optional** and offered to everyone. They had a **quick review in October 2026** by a physiotherapist, a psychologist specialised in pain and a GP: that is not a validation, and a deeper review is still to be done. They are simple scales or answers, **not validated**, and no questionnaire name appears in the app. The report and the Trends card "What else you noted" describe them with the same sentences; nothing is interpreted or compared with a norm. **A day left blank is never counted as "no"**: each share is worked out over the days on which the question was answered ("no" and "none" are answers).

| INDICATOR | CALCULATION | NOTES |
|---|---|---|
| **Morning stiffness** | Days noted, days per band (none, < 30 min, 30 min–1 h, 1–2 h, > 2 h), the band noted most often (none when two tie), mean 0 to 10 intensity when noted. | Bands are OUCH's own, aligned with published reference points for spondyloarthritis; not validated. |
| **Wakings because of pain, woke rested** | Days noted, days with at least one waking, mean per day noted ("3 times or more" counts as 3, which underestimates). Woke rested: days per level (yes, fairly, little, not at all). | Night pain is among the signs described for inflammatory back pain[18]: OUCH describes, it does not classify pain. |
| **Interference with gestures, spasms** | Mean per period, with the previous period, in the symptoms table; "stable" under 0.5 point of change. Sensitivity days rated 0 to 10 before 0.11 still appear there. | Simple scales, not validated. Interference is a single item, inspired by physical functioning, a recommended domain[1]: it is not a questionnaire. |
| **Sensitivity: what bothers** | Four boxes (noise, light, smells, touch). Days with at least one box ticked, over the days where the question was answered (an empty list, "nothing bothers", is an answer); for each box, its number of days. | A reported feeling, **counted in days**: no intensity, no threshold, no comparison. OUCH's own list, not validated; since 0.11 it replaces the 0 to 10 slider. |
| **Painful or swollen joints** | Days with "yes" over the days answered; the three joints named most often (wrist, hand, elbow, knee, ankle, foot, side by side). | A **feeling** the person reports, never an examiner's count. OUCH's own list, independent of the areas in § 10. |
| **Period flow, bleeding outside the period, bowel, urinary** | Days per flow level (none, spotting, light, heavy); for each box, days with "yes" over the days answered. | Sensitive data, optional. Pre-selected by illness at setup, can be unticked with a tap. Nothing is linked beyond what the person notes. |
| **Headache (episode)** | Days answered, days with a headache, days with an attack medication; durations (< 4 h, 4–24 h, > 24 h), days with aura, nausea, bothered by light or noise; the three triggers cited most (case ignored). | The durations are ours, aligned with the 4 to 72 h range of the definition of an attack. OUCH describes an attack, it does not classify it. |
| **Reference point: medication overuse headache** | One sentence restates the definition: headache on at least 15 days a month, regular use for more than 3 months, on at least 15 days a month for simple analgesics and at least 10 days for the other classes[19]. In the report only, and when at least one day has an attack medication. | **No calculation or threshold applied.** Reference points as guideline [19] reports them: the classification itself could not be read at its source. The box does not say the class of the medication. |
| **Leg, signs of the day, fever or infection, new symptom** | Days per leg-pain level (none, down to the knee, below the knee); days with at least one sign ticked and the signs ticked most (five at most); "yes" days for fever or infection and for "new symptom for over 24 hours". | The "new symptom" is the person's own **declaration**: not a relapse of the disease, not a diagnosis, no algorithm reads it. Fever or infection is only context. |

## 12 · COMPARE, NOTE, PREPARE (0.10)

What version 0.10 adds. All of it is **descriptive**: none of these readings is a diagnosis, a score or treatment advice, and none is validated.

| SCREEN | CALCULATION | STATUS |
|---|---|---|
| **Two periods compared (Trends and report)** | The same measures over each period: mean pain, days at 7/10 or more, days inside a detected flare (when flares can be detected), most noted morning-stiffness band (if followed), days logged. There must be **at least 7 logged days on each side**, and the two periods must not overlap. In Trends: the last 30 days and the 30 before, the time since the last appointment and as long before, or two periods of the person's choice. In the report: the report's period and the previous one of the same length, or two periods of the patient's choice. | Description: nothing here says why the two periods differ. |
| **The patient's usual gap** | Consecutive windows of the same length (that of the shorter of the two periods, between 14 and 60 days) are cut from the first logged day. A window counts if at least half its days (7 at least) are logged. The usual gap is the `80% quantile` of the absolute differences in mean pain between two consecutive windows, **from 6 pairs**; otherwise nothing is said. The gap between the two periods is then placed "within" or "beyond" that usual gap. | Same method as section 7. The 80% is the designer's choice; windows are placed end to end and ignore trend and season. |
| **Harder periods noted by the patient (Today, Journal, Trends, report)** | The patient notes the first day, what may have played a part (sleep, stress, effort, weather, period, infection, treatment) and a few words. The period closes when they say so, or by itself after 14 days; 50 at most. **No calculation**: dates and a length. In the report: a table of its own and the mention "+ n period(s) noted by the patient" beside the detected flares, **never added to them**. A period under way lightens the day's entry, like a detected flare. | Declarative: the factors are those the patient mentions, not measured, with no causal hypothesis. Distinct from the definition of flares (section 4). |
| **After the most active days (Trends only)** | If activity is followed and noted on at least 30 days: active days are the upper fifth of the person's activity levels ( `80% quantile` ). A day is "harder than usual" when its pain exceeds the **median of the 28 previous days** (at least 10 logged). Among the active days whose next day is logged (at least 5), those followed by a harder day are counted and compared with what chance would give: the share of harder days over all days of the diary (at least 20 days), applied to the same number of days, with the binomial range from the **10th to the 90th percentile**. The card says "what chance gives", "more than" or "less than". | Descriptive. A day's activity and pain are noted together: **no causality**; days are treated as independent, with no effect of weekday or trend. Thresholds (20%, 28 days, 10th–90th percentile) are the designer's choice; **wording to be reviewed by a health professional**. |
| **Today: the day after a big day** | A quiet message in the "For you" list if yesterday's activity reaches the upper fifth of the last 90 days (at least 20 days logged) and is at least 6/10. Once a day, never during a harder period or a flare; turned off by the "Suggestions" setting. | No result shown, no instruction: a supportive sentence. Absent from the report. |
| **Prescription (Today, Settings)** | A length given by the patient (1 to 24 months) from a day, and a reminder 7 or 14 days before the end, then for 14 days after. **Nothing is worked out from intakes** and no drug is linked. | A reminder, neither advice nor a link to a treatment. Absent from the report. |

## 13 · WHAT VERSION 0.11 CHANGES IN THE REPORT

The first page of both reports is reorganised. Everything stays **descriptive**: nothing in it is a diagnosis, a score or treatment advice, and nothing is validated.

| ELEMENT | CALCULATION | STATUS |
|---|---|---|
| **"In brief" sentence (page 1)** | One sentence built from **the same figures as the tiles** just below it: "Mean pain of *X*/10"; if a previous period exists and its mean is above 0, "down", "up" or "unchanged" by *P*%, with *P* = \|period mean − previous mean\| / previous mean, rounded to the percent ("unchanged" under 0.2 point, as for the arrow); then the number of severe days (≥ 7) and, if flares can be detected and there are some, their number and their days. | It cannot say anything other than the tiles. It describes, it does not conclude; the 30% reference is not in it (see "Key points"). |
| **Days by intensity, compared** | The shares of days ≤ 3, 4–6 and ≥ 7 (§ 3) for the period and for the previous one, as two bars one under the other. For each class, the **gap in percentage points** between the two shares **rounded to the percent** as they are written (34% against 25%: +9 points), so the figures on the page add up. With no previous period, a single bar and no gap. | Green arrow when the change goes the favourable way (more mild days, fewer severe days), orange otherwise; grey for no gap and for the moderate days, which have no "good" side. A description: no test, no threshold. |
| **Trend arrows (tiles)** | The direction of change since the previous period, for mean pain and variability (0–10), the share of severe days, the share of mild days, the share of flare days and sleep duration. A change counts from a gap of **0.2 point** (0–10 scales), **3 percentage points** (shares of days) or **0.25 hour** (sleep); below it, the arrow is a grey dash. | Direction only: not size (the earlier figure is written beside it), nor whether the gap exceeds the patient's usual variation. Green: the favourable direction (pain, severe days and flares down; mild days and sleep up); orange: the opposite. Thresholds set by the designer. |
| **"What happened" list (reports and Trends)** | What touches the period, newest first, by day of start: the flares detected (§ 4), the periods the patient marked themselves, treatment starts, dose changes and stops (§ 7), one-week trials, and the appointments that took place (day, kind, name of the professional; never the note). On the same day: flare, marked period, trial, treatment, then appointment. | A dated list: nothing in it is linked or interpreted. An appointment placed near a dose change does not say that one decided the other. Only the last 6 appointments with the family doctor and the last 2 at the pain clinic are kept. |

### What is not calculated

The contents on page 1 gives the page of each section, measured from the laid-out pages. The orientation texts of the report (the numbered parts of the pain clinic report) contain no calculation. The length announced to the patient (3 to 5 pages, 6 to 10 pages) is a range, not a measurement.

## APPENDIX · WHAT THE APP SHOWS THE PERSON, OUTSIDE THE REPORT

| SCREEN | CALCULATION | STATUS |
|---|---|---|
| **Trends: leads** | For sleep (the value entered that day), stress, the previous day's activity and each action noted under "what helped": at least 14 days, a Spearman correlation (section 6) of at least $\max(0.3; 2/\sqrt{n})$, two groups of at least 4 days (upper and lower third of the person's values, or days with and without the action) and a point of mean pain between them. At most three leads, the strongest first, those the person kept coming first. Fatigue, brain fog, mood and weather are not among them. | Same correlation threshold as the report; 4 days and 1 point chosen for readability, unsourced. No correction for multiple comparisons: with several factors, a false lead is possible. Presented as "a lead, not proof". |
| **Leads: the patient's answers** | "Yes, note it for my appointment" keeps the lead; "Not really" sets it aside. A lead set aside comes back if its gap grows by a point, if the number of days doubles or if it changes direction. | Preferences: they change no calculation. Only kept leads go on the report (section 9). |
| **Today: "for you", a dot** | A line under the title ("2 small things for you") opens the list of what waits: leads, the Sunday sentence, a trial to repeat, the invitation. "Later" sets a message aside for 7 days; at the second, it no longer comes back. No line during a flare. The dot flags a lead not shown yet. | Display: no further calculation. "Suggestions" turns off the leads, the Sunday sentence, the trials and the dot. |
| **Your week in one sentence** | On Sundays, if at least 5 days of the week are logged and the 3 gentlest days share one action (present on at most one other day) or a declared night of at least 7 h. No tie for third place, no flare under way. | Descriptive, on three days: no test, to be read as a remark. |
| **One-week trial** | After "Yes" on an action lead: 7 days, mean pain on the days with the action and on the other days, if each kind has at least 2 days, and whether it goes the same way as the lead it started from. The action is the one the patient noted themselves. | A description of seven days, presented as "an idea, not an answer": no test, and no control for what else changes in the week. |
| **Trends: day of the week** | Best and worst days, if each weekday has ≥ 4 days and one of the two falls outside the band (section 3); otherwise, a sentence says the days look alike. | Descriptive. |
| **Trends: before and after a change** | Same calculation as in section 7, plus sleep and the effects reported before and after, and a curve on request. Each measure is placed "within" or "beyond" the person's usual variation, when the history allows. | Description, presented as such: it does not say whether the treatment "works". Replaces the former 2-week comparison. |
| **Today, on flare days** | Pain is rated in eleven boxes to tap (0 to 10), with "one point less / more" to correct. Nothing is preselected, so as not to anchor the answer on a starting value. A single blue, deeper as pain rises. | Same 0–10 scale: entries and report unchanged. No source cited on the effect of this input mode. |
| **Weather of the day (shareable card)** | Score $0.60 \times \text{pain} + 0.25 \times \text{fatigue} + 0.15 \times \text{brain fog}$, renormalised if one of the last two is missing. Five levels, from "clear" to "storm": < 1.5, < 3.5, < 5.5, < 7.5, ≥ 7.5. | **Weights chosen by the designer, not validated.** Image to share with loved ones, absent from the report. |
| **Offer to log yesterday** | In the morning, if yesterday has no pain rating while the day before has one: a line in the day's list, set aside with one tap, never during a flare under way. Recall is more faithful the next day than a few days later[6][22]. No calculation depends on it. | |
| **"The last few days have been harder" card** | Shown once per flare under way the day before, with what the person noted as helpful during their previous flares. During the flare, entry shrinks to pain, the rest on request. | No outside suggestion; both can be turned off. |

## APPENDIX (CONTINUED) · PROPOSED INDICATORS AND APPOINTMENT

| SCREEN | CALCULATION | STATUS |
|---|---|---|
| **Setup and Settings: proposed indicators** | A catalogue links each indicator to the declared illnesses: recommended, suggested, or free choice. Union of the profiles of every illness ticked (“diagnosed” or “suspected”: same profile); a cap of 7 visible measures, pain included. Beyond it, what defines an illness (migraine episode, period…) comes before the shared measures, then the order of priority. With no illness, or “other” alone: pain, fatigue, sleep, mood, areas, medications, notes. | **Profiles proposed by the designer**, from the sources cited in § 11: to be reviewed. It is a sort, not a prescription: no indicator is out of reach. |
| **Setup: what is ticked** | On the first pass only, the recommended ones replace the eleven factors ticked by default. Going back to the step, running the setup again or changing illness afterwards changes no choice: what becomes recommended is shown unticked. | A setting, no calculation on the data. Cycle tracking starts on with endometriosis. |
| **Psoriatic arthritis** | Indicator profile: the stronger of the two tiers of rheumatoid arthritis and spondyloarthritis, for each indicator. | No source of its own has been read for it. Early access only. |
| **Invitation (existing users)** | A message in the “For you” list when recommended or suggested indicators are not followed. Never during a flare. “Later” postpones it by 7 days; the second one is the last. “Add” adds only what is ticked and removes nothing. | Display and preferences: no further calculation. |
| **Day’s page: words of recognition** | Three messages, with no score or streak: a sentence at the end of a one-week trial (if the action was noted at least one day); “Back to calm” within 7 days after a flare that is over, whose return to the usual level is known (≥ 1 day), with that number of days; and, in the first 7 days of the month, a summary of the month gone by (≥ 10 logged days, ≥ 2 lines): trials seen through, the softest week of seven days from the 1st (≥ 5 logged days, at least two comparable weeks), the average days to come back to the usual level after the flares. Never during a flare; “Thanks” closes them for good; they follow the “Suggestions” setting. | Descriptive, with no well-being threshold or comparison with other people. Not in the report. |
| **Trends: “What else you noted”** | The sentences of § 11 over the chosen period, one line per indicator; the card appears only when a day holds one. Without the attack-medication reference point, which is in the report only. | Descriptive, no threshold or comparison. |
| **Appointment, questions and note (Today, Trends, report)** | The patient notes the date, their questions, what they most want to talk about and, after the appointment, a note on what was said. **No calculation.** In the report: the “What the patient wishes to raise” list, the note (the patient can leave it out), and the report’s period starts by default at the last appointment’s date. The sheet also recalls, since that appointment, mean pain, harder periods and treatment changes. | Free text from the patient, reproduced as written. The recalled figures are those of sections 3, 4 and 7. |

## REFERENCES, AND WHAT EACH ONE CONTRIBUTES

Each summary restates the main conclusion of the study, from its official abstract (PubMed); it is the same text as in the app's "Scientific sources" section. These studies concern groups of patients: they establish nothing certain for a given person.


1. Dworkin RH, Turk DC, Farrar JT, et al. Core outcome measures for chronic pain clinical trials: IMMPACT recommendations. *Pain*. 2005;113(1-2):9-19. PMID 15621359.

   **What the study says.** Expert consensus (IMMPACT) on the measures to include in clinical trials of chronic pain: pain intensity, rated on a 0 to 10 numeric scale, is one of them, along with functioning, mood and the global impression of improvement. **In OUCH.** Pain is rated each day from 0 to 10, as in clinical trials, which makes the figures readable by a clinician.

2. Farrar JT, Young JP Jr, LaMoreaux L, Werth JL, Poole MR. Clinical importance of changes in chronic pain intensity measured on an 11-point numerical pain rating scale. *Pain*. 2001;94(2):149-158. PMID 11690728.

   **What the study says.** Across 2,724 patients in 10 clinical trials, a drop of about 2 points, or about 30%, on the 0–10 scale corresponds on average to an improvement the patient judged important. A larger drop is needed when baseline pain is high. **In OUCH.** The report states whether the change in pain reaches this 30% threshold. The same benchmark (2 points or 30%) is used to detect flares, as a rise: this is an extrapolation, established for drops. It is a group average: it does not say what matters to the person.

3. Haute Autorité de santé. Douleur chronique : reconnaître le syndrome douloureux chronique, l'évaluer et orienter le patient. Recommandations professionnelles, décembre 2008 (in French).

   **What the study says.** French guideline that defines chronic pain syndrome (persistent or recurrent pain with functional impact) and describes how to recognise it, assess it and refer the patient. **In OUCH.** French reference cited in the report for the assessment of chronic pain.

4. Dworkin RH, Turk DC, Wyrwich KW, et al. Interpreting the clinical importance of treatment outcomes in chronic pain clinical trials: IMMPACT recommendations. *J Pain*. 2008;9(2):105-121. PMID 18055266.

   **What the study says.** Consensus of 40 participants (universities, agencies, patients, industry) on interpreting results: at least two methods are advised, and provisional benchmarks (about 30% drop: moderate; about 50%: substantial) are proposed. **In OUCH.** The report describes a 30% drop as moderately relevant and 50% as substantial. These benchmarks are designed for groups of patients, not for an individual.

5. Boonstra AM, Stewart RE, Köke AJA, et al. Cut-off points for mild, moderate, and severe pain on the numeric rating scale for pain in patients with chronic musculoskeletal pain. *Front Psychol*. 2016;7:1466. PMID 27746750.

   **What the study says.** Across 2,854 chronic pain patients, the cut-offs depend on the impact on daily life: on average, mild ≤ 5, moderate 6–7, severe ≥ 8; among those who catastrophise little, mild ≤ 3, moderate 4–6, severe ≥ 7. **In OUCH.** OUCH uses mild ≤ 3, moderate 4–6, severe ≥ 7, conventional thresholds that match this second case. They are benchmarks, not a diagnosis.

6. Broderick JE, Schwartz JE, Vikingstad G, et al. The accuracy of pain and fatigue items across different reporting periods. *Pain*. 2008;139(1):146-157. PMID 18455312.

   **What the study says.** In 83 patients followed with an electronic diary, recalled pain is inflated compared with ratings made through the day. Recall of the previous day remains faithful, but memory degrades beyond a few days. **In OUCH.** OUCH has the person rate the day just ended, day by day, rather than summarise the week from memory.

7. Dixon WG, Beukenhorst AL, Yimer BB, et al. How the weather affects the pain of citizen scientists using a smartphone app. *NPJ Digit Med*. 2019;2:105. PMID 31667359.

   **What the study says.** "Cloudy with a Chance of Pain": 2,658 patients rated their pain every day for 15 months. Links exist with humidity, atmospheric pressure and wind, but they are modest. **In OUCH.** OUCH records the day's weather so the person can see whether it matters for them, without presenting the link as established.

8. Salaffi F, Stancati A, Silvestri CA, Ciapetti A, Grassi W. Minimal clinically important changes in chronic musculoskeletal pain intensity measured on a numerical rating scale. *Eur J Pain*. 2004;8(4):283-291. PMID 15207508.

   **What the study says.** In 825 patients, a drop of one point (15%) is the smallest perceptible difference; a drop of 2 points (33%) corresponds to "much better". A larger drop is needed when baseline pain is high. **In OUCH.** Confirms the 2-point or 30% benchmark used by the report.

9. Diener HC, Kropp P, Dresler T, Evers S, Förderreuther S, Gaul C, Holle-Lee D, May A, Niederberger U, Moll S, Schankin C, Lampl C. Management of medication overuse (MO) and medication overuse headache (MOH) S1 guideline. *Neurol Res Pract*. 2022;4(1):37. PMID 36031642.

   **What the study says.** A guideline from the German, Austrian and Swiss neurology societies. It recalls the definition of medication overuse headache in the international classification of headache disorders: headache on at least 15 days a month in someone who already has a headache disorder, and regular use of an attack medication for more than 3 months, on at least 15 days a month for simple analgesics and at least 10 days a month for combination analgesics, triptans, ergot derivatives and opioids. **In OUCH.** For someone who notes days with an attack medication, the report restates these reference points in one sentence, with their source, next to the count. OUCH does not know the class of the medication (it is a box ticked), computes nothing and does not conclude: only a doctor can say a person is concerned. The classification itself could not be read at its source; these are the reference points as this guideline reports them.




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## REFERENCES, CONTINUED


9. Ostelo RWJG, Deyo RA, Stratford P, et al. Interpreting change scores for pain and functional status in low back pain: towards international consensus regarding minimal important change. *Spine*. 2008;33(1):90-94. PMID 18165753.
   **What the study says.** Expert consensus for low back pain: a 2-point change on the 0–10 scale is a minimal important change, and a 30% improvement is a useful threshold whatever the scale. **In OUCH.** Confirms, by another route, the 2-point or 30% benchmark.

10. Hawker GA, Mian S, Kendzerska T, French M. Measures of adult pain. *Arthritis Care Res*. 2011;63(S11):S240-S252. PMID 22588748.
    **What the study says.** Review of adult pain measurement tools, including the numeric scale and the visual analogue scale, and their measurement properties. **In OUCH.** One of the sources that justify the 0–10 numeric scale for rating pain.

11. Beukenhorst AL, Schultz DM, McBeth J, Sergeant JC, Dixon WG. Are weather conditions associated with chronic musculoskeletal pain? Review of results and methodologies. *Pain*. 2020;161(4):668-683. PMID 32195783.
    **What the study says.** Review of 43 studies: two in three find a link between weather and pain, but in varying directions, often in small groups followed for less than a month. The authors call for larger and longer studies. **In OUCH.** This is why the link with atmospheric pressure is presented as “inconsistent literature” in the report.

12. Smedslund G, Hagen KB. Does rain really cause pain? A systematic review of the associations between weather factors and severity of pain in people with rheumatoid arthritis. *Eur J Pain*. 2011;15(1):5-10. PMID 20570193.
    **What the study says.** Review of 9 studies in rheumatoid arthritis: on average, the link between pain and temperature, humidity or pressure is close to zero, but fewer than one patient in four reacts to it individually. **In OUCH.** Reaction to weather varies from person to person: OUCH looks at the person’s own data rather than the average.

13. Steffens D, Maher CG, Li Q, et al. Effect of weather on back pain: results from a case-crossover study. *Arthritis Care Res*. 2014;66(12):1867-1872. PMID 25044376.
    **What the study says.** In 993 patients, temperature, humidity, pressure and rain do not increase the risk of acute low back pain. Strong wind increases it slightly, without clinical importance. **In OUCH.** Another reason to present the weather–pain link only as a lead to discuss.

14. Finan PH, Goodin BR, Smith MT. The association of sleep and pain: an update and a path forward. *J Pain*. 2013;14(12):1539-1552. PMID 24290442.
    **What the study says.** Review: sleep disturbances predict the onset and worsening of chronic pain, and predict pain more strongly than pain predicts sleep. The link nonetheless runs in both directions. **In OUCH.** The report pairs pain with the previous night’s sleep, noting that the link runs both ways.

15. Andrews NE, Strong J, Meredith PJ. Activity pacing, avoidance, endurance, and associations with patient functioning in chronic pain: a systematic review and meta-analysis. *Arch Phys Med Rehabil*. 2012;93(11):2109-2121. PMID 22728699.
    **What the study says.** Review of 41 studies: avoiding activity and pushing through to severe pain are both associated with more pain and disability. The direction of causation is not established. **In OUCH.** The report compares pain the day after a very active day, noting that too much and too little activity are both associated with more pain.

16. Wolfe F, Clauw DJ, Fitzcharles MA, Goldenberg DL, Katz RS, Mease P, Russell AS, Russell IJ, Winfield JB, Yunus MB. The American College of Rheumatology preliminary diagnostic criteria for fibromyalgia and measurement of symptom severity. *Arthritis Care Res*. 2010;62(5):600-610. PMID 20461783.
    **What the study says.** Preliminary American College of Rheumatology criteria for fibromyalgia, which no longer require a tender-point examination. They combine a widespread pain index (WPI), the number of body regions, out of 19, where the person had pain, and a symptom severity scale. **In OUCH.** The areas of “Where does it hurt?” follow the 19 WPI regions, with the left or right side, plus head, hands and feet. OUCH does not compute the WPI, which covers the past week and is for the doctor: it counts, day by day, what the person reports.

17. Wolfe F, Clauw DJ, Fitzcharles MA, Goldenberg DL, Häuser W, Katz RL, Mease PJ, Russell AS, Russell IJ, Walitt B. 2016 Revisions to the 2010/2011 fibromyalgia diagnostic criteria. *Semin Arthritis Rheum*. 2016;46(3):319-329. PMID 27916278.
    **What the study says.** A revision of those criteria after a review of 14 validation studies. The WPI and its 19 regions are kept, and generalized pain requires pain in at least 4 of the 5 main body regions (jaw, chest and abdomen excluded). **In OUCH.** The same split of the body. The report says how many of those regions are reported over the period, as a plain count. OUCH applies neither these criteria nor the definition of generalized pain: only a doctor makes a diagnosis.

18. Sieper J, van der Heijde D, Landewé R, Brandt J, Burgos-Vagas R, Collantes-Estevez E, Dijkmans B, Dougados M, Khan MA, Leirisalo-Repo M, van der Linden S, Maksymowych WP, Mielants H, Olivieri I, Rudwaleit M. New criteria for inflammatory back pain in patients with chronic back pain: a real patient exercise by experts from the Assessment of SpondyloArthritis international Society (ASAS). *Ann Rheum Dis*. 2009;68(6):784-788. PMID 19147614.
    **What the study says.** Thirteen expert rheumatologists examined 20 people with back pain. Five signs weigh most in their judgement of inflammatory pain: relief with exercise, pain at night, a gradual start, a start before age 40 and no relief with rest. With at least four of the five signs, sensitivity was 77.0% and specificity 91.7% in that group, then 79.6% and 72.4% in a validation group of 648 people. **In OUCH.** It supports the “wakings because of pain” and “what helped” (movement or rest). OUCH describes, it does not classify pain or count any criterion: only a doctor makes a diagnosis.

## REFERENCES, CONTINUED


20. Broderick JE, Schwartz JE, Schneider S, Stone AA. Can end-of-day reports replace momentary assessment of pain and fatigue? *J Pain*. 2009;10(3):274-281. PMID 19070550.

    **What the study says.** In 105 rheumatology patients, the evening rating for the whole day is very close to the mean of several ratings taken through the day (r = 0.90 to 0.92 for pain, lower for fatigue). **In OUCH.** One evening rating is enough for the day's average level.

21. Schneider S, Stone AA, Schwartz JE, Broderick JE. Peak and end effects in patients' daily recall of pain and fatigue: a within-subjects analysis. *J Pain*. 2011;12(2):228-235. PMID 20817615.

    **What the study says.** Summing up a day from memory gives slightly more weight to the strongest moment and the end of the day: a small effect for pain, none for fatigue. **In OUCH.** The screen says "over your whole day, not just right now".

22. Marty M, Rozenberg S, Legout V, et al. Influence of time, activities, and memory on the assessment of chronic low back pain intensity. *Spine*. 2009;34(15):1604-1609. PMID 19564771.

    **What the study says.** In 194 patients with low back pain followed for 29 days: the evening rating for the day is accurate, but pain recalled 24 or 48 h later is not, and pain recalled over 7 or 28 days mostly follows the pain of the day of answering. **In OUCH.** A reason to rate in the evening or the next morning; more cautious than [6] about the day before, hence the two-day threshold in the report.

23. Carlozzi NE, Schilling S, Freedman J, Kalpakjian CZ, Kratz AL. The reliability of end of day and ecological momentary assessments of pain and pain interference in individuals with spinal cord injury. *Qual Life Res*. 2018;27(11):3003-3012. PMID 30073468.

    **What the study says.** Followed for 7 days, the time of the measures (morning, midday, evening) did not change the estimated reliability of pain; their number does. **In OUCH.** No ideal time is demonstrated: the reminder time is free to set.




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